No.	Status	Date
QSCP-NDT-001 Rev 1	Issued	2008-04-01

Site NDE Procedure - Ultrasonic Thickness Measurement, ATU-100

1.0 Scope and Purpose

This procedure establishes the requirements for ultrasonic thickness measurement within the ATU-100. It supersedes QSCP-NDT-001 Rev 0 to incorporate critical technical updates following the 2-RV-102 elbow leak. The primary objective is to ensure accurate detection of localized corrosion in acid gas service, specifically addressing the limitations identified in previous inspection methodologies.

The scope includes all piping circuits defined in QSCP-IP-ATU100 Rev 1, with specific focus on the regenerator overhead circuit. This revision mandates the use of digital ultrasonic gauges equipped with A-scan waveform displays to verify signal integrity and differentiate between true wall thinning and laminar inclusions or signal loss.

2.0 Responsibilities

Douglas Ferreira is responsible for the technical oversight of NDE activities, ensuring that all field technicians are qualified and that data collection aligns with the revised CML requirements. The Inspection Supervisor must verify that all readings below t-min are immediately documented as an NCR per site rules.

Gordon Lisle is responsible for the final review of inspection data and the approval of calculated remaining life and next inspection dates. The Unit Engineer ATU-100 ensures that the inspection intervals for the regenerator overhead circuit do not exceed the 3-year limit established in the revised piping inspection plan.

3.0 Reference Documents

This procedure implements the corrective actions defined in QSCP-RCA-2007-002, which identified localized corrosion from condensing acid gas as the cause of the 2-RV-102 failure. The analysis determined that spot readings and the absence of CMLs on fittings were significant contributing factors to the undetected wall loss.

Inspection frequencies and CML locations are governed by QSCP-IP-ATU100 Rev 1. This document reflects the addition of 20 fitting CMLs, bringing the unit total to 138, and accounts for the replacement of the failed elbow identified during the root cause investigation.

4.0 Inspection Methodology

Technicians shall perform continuous scanning rather than spot readings at all elbows and tees. This scanning technique is mandatory to capture localized thinning at fittings, as required by the findings in QSCP-RCA-2007-002. All measurements must be performed using gauges with waveform display to ensure the accuracy of the backwall echo.

CMLs must be placed directly on fittings as specified in the updated piping plan. For the regenerator overhead circuit, which is classified as Class 1, the thickness inspection interval is set at 3 years. All other Class 1 circuits follow a 5-year maximum interval, while Class 2 and Class 3 follow 10-year maximums.

5.0 Data Evaluation and Remaining Life

The corrosion rate for each CML shall be the larger of the long-term rate and the short-term rate, provided the latest pair of readings are at least 3 years apart. Remaining life is calculated by dividing the difference between the actual thickness and t-min by the determined corrosion rate. The next inspection date must be scheduled at no more than half the remaining life.

Any reading recorded below the line t-min requires the immediate issuance of an NCR. Douglas Ferreira shall review these findings to determine if additional scanning is required to delineate the boundaries of the thinned area. Gordon Lisle will provide final engineering disposition for any piping components falling below retirement thickness.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2008-04-01	_____
Approved	Gordon Lisle	Unit Engineer ATU-100, P.Eng.	2008-04-01	_____